FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2021

PAPER –2

TEST DATE: 30-05-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. BC

Sol. P_{max} at resonance $\Rightarrow \omega = \frac{1}{\sqrt{LC}}$

$$f = \frac{1}{2\pi}\sqrt{10\times 40\times 10^{-6}} = \frac{25}{\pi} Hz$$

$$Q_{factor} = \frac{\omega}{2\Delta\omega} = \frac{\omega L}{R} \Rightarrow \Delta\omega = \frac{R}{2L} = \frac{100}{20} = 5$$

$$\Delta f = \frac{45}{2\pi}$$

2. AD

3. ABCD

Sol. By the graph it is clear the process is isothermal.

$\therefore P_A V_A = P_B V_B$

(A) and (B) are correct

Now by the ideal gas equation.

$$PV = nRT \Rightarrow \frac{PM}{\rho} = nRT$$

$$P = \left(\frac{nRT}{M}\right)\rho$$

$$P \propto \rho$$

$\therefore$ from A to B pressure is increasing and volume decreasing.

4. AD

Sol. $\tau_{mg} = 0.5g \times 1\hat{i} = 5\hat{i}$

$= \tau_B = \left(-0.5 \times 4\hat{i}\right) \times \vec{B}$

Option A $\tau_B = -0.5\hat{i} + 6\hat{k}$ those $\vec{B}$ for which $\vec{\tau}$ has $-5\hat{i}$ are useful.

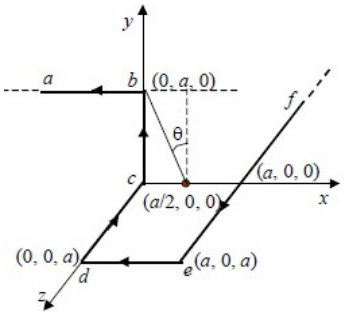
5. ABCD

Sol. $\vec{B}_{ab} = \dfrac{\mu_0 I}{4\pi a}\left[\sin\left(-\theta_1\right) + \sin\dfrac{\pi}{2}\right]\left(\hat{k}\right) = \dfrac{\mu_0 I}{4\pi a}\left[1 - \dfrac{1}{\sqrt{5}}\right]\hat{k}$

$\vec{B}_{bc} = \dfrac{\mu_0 I}{4\pi.\dfrac{a}{2}}\left[\sin\theta + \sin\left(90 - \theta\right)\right] = \dfrac{\mu_0 I}{2\pi a}\left[\dfrac{2}{\sqrt{5}}\right]\left(-\hat{k}\right) = \dfrac{-\mu_0 I}{\pi a\sqrt{5}}\hat{k}$

$\vec{B}_{cd} = \dfrac{-\mu_0 I}{\pi a\sqrt{5}}\, j \qquad \left[\left|\vec{B}_{cd}\right| = \left|\vec{B}_{bc}\right|\right]$

$\vec{B}_{ef} = \dfrac{-\mu_0 I}{2\pi a}\left[1 + \dfrac{2}{\sqrt{5}}\right]\hat{j}$



6. AD

Sol. $nR = \dfrac{1}{25} \times 10^2 = 4 \Rightarrow \dfrac{m}{32} = \dfrac{4}{25} \times 3$

$m = \dfrac{96 \times 4}{25} = 15.36gm$

$V = \dfrac{nRT}{p}$

$V_{\max} at\, 3$

$nR = \dfrac{16 \times 40}{400}$

$\mathrm{V}_1 = \dfrac{1}{25} \times \dfrac{300}{1}$

$\mathrm{V}_2 = \dfrac{1}{25} \times \dfrac{400}{2}$

SECTION – B

7. 2

Sol. $\mathrm{P}_{ab} = \dfrac{\text{heat energy}}{\text{time}}$

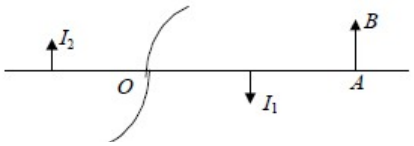
$\left(\text{Intensity}\right)\left(\text{cross section area}\right) = \dfrac{\mathrm{ms}\Delta\mathrm{T}}{\mathrm{t}}$

$t = \dfrac{\left(4 \times 10^{-3}\right)\left(4200\right)\left(5\right)}{\left(1000\right)\left(1.4 \times 10^{-4}\right)} = 600s = 10\,\text{min}$

8. 6

Sol. $V_1 = V_2$

$$\frac{(-x)\times(-10)}{-x+10} = \frac{x\times 15}{-x-15} \Rightarrow x = 60cm$$



9. 7

Sol. $$\frac{1}{5} = 1 - \frac{Q_L}{Q_g} \Rightarrow Q_L = \frac{4}{5}Q_g$$

$$\frac{1}{10} = 1 - \frac{Q'_L}{Q_L}$$

$$Q'_L = \frac{9}{10}Q_L = \frac{36}{50}Q_g$$

$$\eta = 1 - \frac{Q'_L}{Q_g} = 1 - \frac{36}{50} = \frac{14}{15} \Rightarrow 28\%$$

10. 2

11. 2

12. 3

SECTION – C

13. 00006.00

14. 00005.00

15. 00002.20

Sol. $$t = \frac{T}{2} = \frac{\pi m}{qB} = \frac{\left(\frac{22}{7}\right)\times\left(14\times10^{-3}\right)}{\left(3\times10^{-6}\right)\left(200/3\right)} = 220s$$

16. 00006.00

Sol. $$R = \frac{2T}{\rho gh} = \frac{2\times73\times10^{-3}}{10^3\times9.8\times25\times10^{-3}} \qquad = 0.6\times10^{-3}m > 0.25mm$$

17. 00001.00

Sol. $$M = \sqrt{2}\times i.\frac{\pi r^2}{2} \qquad = \sqrt{2}\times i.\frac{\pi}{2}\times\left(\frac{l}{2\pi}\right)^2 \quad = \frac{\sqrt{2}iL^2}{8\pi}$$

18. 00005.00

Sol. $$y = 8\sin\left\{\pi\left(f_1 - f_2\right).t\right\}\cos\left\{\pi\left(f_1 + f_2\right)t\right\}$$

Chemistry

PART – II

SECTION – A

19. ACD

20. ACD

21. AC

22. ABCD

23. ACD

24. CD

SECTION – B

25. 3
Sol. As x=7, and y= 4

26. 5

27. 5

28. 5

29. 9
Sol. except 10 all are correct

30. 6
Sol. n can be 0, 1, 2 or 3

SECTION – C

31. 00067.50

Sol. Fraction of one product $= \frac{1\times10^{-2}}{1\times10^{-2}+3\times10^{-2}} = 0.25$

Fraction of another product $= \frac{3\times10^{-2}}{1\times10^{-2}+3\times10^{-2}} = 0.75$

Overall activation energy is

$$= \frac{0.25\times60+0.75\times70}{0.25+0.75} = 67.5\,kJ/mol$$

32. 00000.83

33. 00000.26

Sol.

$$CH_3OCH_3 \longrightarrow CH_4 + CO + H_2$$

$t=0$ 0.4

$t=4.5\,hr.$ 0.11 0.29 0.29 0.29

$$kt = \ln\frac{p_0}{p} \Rightarrow 4.78\times10^{-3}\times4.5\times60 = \ln\frac{0.4}{p}$$

$p = 0.11 atm$

At $t = 0 \quad p_0 = 0.4$

$M_0 = 46$

At $t = 4.5 hr. \; p = 0.11 + 0.29 \times 3 = 0.98 atm$

$$M = \frac{0.11 \times 46 + 0.29(16 + 28 + 2)}{0.98} = 18.77$$

$$\frac{R_0}{R} = \frac{p_0}{p}\sqrt{\frac{M}{M_0}} = \frac{0.4}{0.98}\sqrt{\frac{18.77}{46}} = 0.26$$

34. 00005.00

Sol. $$\frac{p° - p_s}{p_s} \times \frac{1000}{m_A} = m = \frac{\Delta T_f}{K_f}$$

$$\therefore \text{m} = \frac{100 - 98.88}{98.88} \times \frac{1000}{78} = 0.1452$$

Now, $K_f = \frac{\Delta T_f}{m} = \frac{0.73}{0.1452} = 5.027$

35. 00000.47

Sol. Miliequivalents of NH_3 formed

= Miliequivalents of HCl used

$= 100 \times 0.15 - 32.10 \times 0.10$

$= 11.79$

Here, n-factor of NH_3 is 1 (acid – base reaction)

$\overset{+5}{\text{NO}_3^-} \xrightarrow{+8e^-} \overset{-3}{NH_3} \quad (n - factor = 8)$

$\therefore$ Miliequivalents of NH_3 for n-factor 8 = 8× 11.79 = 94.32

$\therefore$ normality of $\text{NO}_3^- = \frac{\text{miliequivalent}}{\text{V(ml)}} = \frac{94.32}{25} = 3.77$

$$\text{Molarity of } \text{NO}_3^- = \frac{3.77}{8} = 0.47125$$

36. 00010.59

Sol. $\text{C}_\text{p} = 22.34 + 48.1 \times 10^{-3} T \qquad JK^{-1} mol^{-1}$

$$\Delta H = \int^{n=1} nC_p dT = \int_{T_1}^{T_2} \left(22.34 + 48.1 \times 10^{-3} T\right)$$

$$\Delta H = 22.34 \times 298 + \frac{48.1 \times 10^{-3} \times 298 \times 3}{2} \times 298$$

$\Delta H = 13.064 kJ / mole$

$W = -P(V_2 - V_1) = -nRT(T_2 - T_1)$

$W = -2477 J = -2.477 kJ$

$\Delta E = 10.587 kJ$

Mathematics

PART – III

SECTION – A

37. BC

Sol. $f(x) = a^2x^2 - 8ax + 7$

$f(1) < 0$

$(a-7)(a-1) < 0$

$1 < a < 7$

38. AB

Sol. $w = \dfrac{z-1}{z-2}$

$z = \dfrac{2w-1}{w-1}$

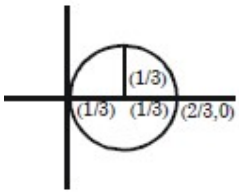
$|z| = 1$

$\Rightarrow |w-1|(\overline{w}-1) = (2w-1)(2\overline{w}-1)$

$3|w|^2 - w - \overline{w} = 0$

$3(x^2+y^2) - 2x = 0$

$\left(x - \dfrac{1}{3}\right)^2 + y^2 = \dfrac{1}{9}$



39. ABC

Sol. We know that sgn x = $\begin{cases} 1 & x > 0 \\ -1 & x < 0 \\ 0 & x = 0 \end{cases}$ and $\{\sin x\} \in [0, 1)$

$\Rightarrow$ f(x) = $\begin{cases} 1 & -1 < \sin x < 1, \sin x \neq 0 \\ 0 & \sin x = -1, 0, 1 \end{cases}$

f(x) is a periodic function with period π and f(–x) = $\begin{cases} 1 & -1 < \sin x < 1, \sin x \neq 0 \\ 0 & \sin x = -1, 0, 1 \end{cases}$ = f(x)

being a periodic function, it is many to one.

40. AC

Sol. (A) In [2007, 2009]

$(\ell n\ x)^{2007} < (\ell n\ x)^{2009}$

Hence $\int_{2007}^{2009} (\ell n\ x)^{2007}\ dx < \int_{2007}^{2009} (\ell n\ x)^{2009}\ dx$

(B) In (0, 1)
$x f(x) < f(x)$
as $f(x) > 0$

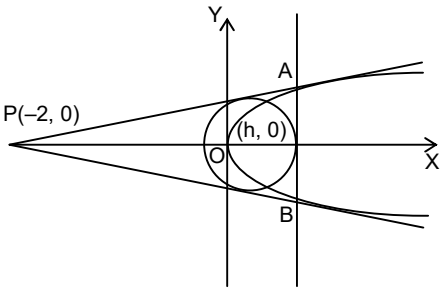
(C) $\int_1^{\frac{1}{x}} \frac{dt}{1+t^2}$ Put $t = \frac{1}{z}$ $\quad -\int_1^x \frac{dz}{1+z^2} = \int_x^1 \frac{dz}{1+z^2}$

(D) $\int_0^\infty e^{-x^2} dx = \int_0^\infty \frac{e^{-t}}{2\sqrt{t}}\ dt$

41. ABCD

42. AB

Sol. Point P clearly lies on the directrix of $y^2 = 8x$.
Thus, slopes of PA and PB are 1 and –1 respectively.
Equation of PA : $y = x + 2$,
Equation of PB : $y = -x - 2$, Equation of AB : $x - 2$
Let the centre of the circle be (h, 0) and radius be r.



$\Rightarrow \frac{|h+2|}{\sqrt{2}} = \frac{|h-2|}{1} = r \Rightarrow h^2 + 4 + 4h = 2(h^2 + 4 - 4h)$

$\Rightarrow h^2 - 12h + 4 = 0$

$\therefore h = \frac{12 \pm 8\sqrt{2}}{2} = 6 \pm 4\sqrt{2}$

$\Rightarrow$ radius $= |h - 2| = 4\left(\sqrt{2} - 1\right), 4\left(\sqrt{2} + 1\right)$.

SECTION – B

43. 0

44. 1

Sol. $f'(x) = 3x^2 + 6x + 4 - k \geq 0\ \forall x \in R$

$D \leq 0$

$36 - 12(4 - k) \leq 0$

$3 - 4 + k \leq 0$

$\therefore k \leq 1$

$\Rightarrow k_{\max} = 1$

45. 3

Sol. $s = \frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \frac{7}{2^4} + ...\infty$

$\frac{s}{2} = \frac{1}{2^2} + \frac{3}{2^3}\frac{7}{2^4} + ...\infty$

$\frac{s}{2} = \frac{1}{2}\left\{\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + ...\infty\right\}$

$s = 3$

46. 0

Sol. $a+b+c=0$

$\Rightarrow \bar{a}+\bar{b}+\bar{c}=0$

$\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$

$ab+bc+ca=0$

$\Rightarrow a^2+b^2+c^2=0$

47. 4

Sol. $S = \frac{16+3x+5x}{2} = 8 + 4x,$

Now $3x + 5x > 16 \Rightarrow x > 2$

$3x + 16 > 5x \Rightarrow x < 8 \Rightarrow x \in (2, 8)$

$5x + 16 > 3x \Rightarrow x > -8$

Now $A^2(x) = (8 + 4x)(4x - 18)(x + 8)(8 - x) = (16x^2 - 64)(64 - x^2)$

Let $x^2 = t \Rightarrow t \in (4, 64)$, $f(t) = 16(t - 4)(64 - t) = 16(64t - t^2 - 256 + 4t)$

$\therefore f(t) = -16(t^2 - 68t + 256)$, $f'(t) = 2t - 68 = 0 \Rightarrow t = 34$

$f''(t) = -32 < 0$

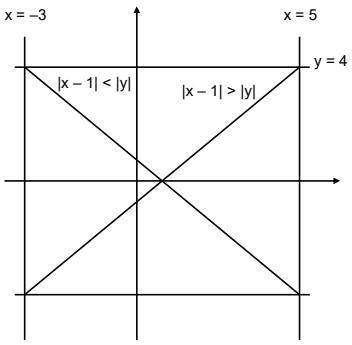
Maxima occurs at $t = 34$

$F(34) = 16(30)(30)$

$\therefore$ largest possible area = 120.

48. 6

Sol. Required area = $8 \times 8 = 64$.



SECTION – C

49. 00003.00

Sol. $x^{14}+x^2+2=kx^4$

$x^{10}+\frac{2}{x^4}+\frac{1}{x^2}=k$

$\frac{x^{10}+\frac{1}{x^4}+\frac{1}{x^4}+\frac{1}{x^2}}{4} \geq 1$

$\Rightarrow$ for $k < 4$ solution

$k_{max} = 3$

50. 00008.56

51. 00000.67

Sol. since, $\frac{\tan\theta}{\tan\theta-\tan 3\theta}+\frac{\cot\theta}{\cot\theta-\cot 3\theta}=1$

52. 00036.00

Sol. On solving we get point of intersection as $(-12,\ 24,\ -24)$ whose distance from origin is 36.

53. 00001.62

Sol. $y=\frac{x}{x^2+y}$

$\Rightarrow y^2+x^2y-x=0$

$\therefore \frac{1}{y}\frac{dy}{dx}=\frac{-\left(x^2-y\right)}{\left(x^2+y\right)\left(x+y^2\right)}$

$\Rightarrow \log y+c=\int\frac{\left(y-x^2\right)dx}{\left(x^2+y\right)\left(x+y^2\right)}$

$\therefore f'(y)=\frac{1}{y}$ and for $x=1,\ y^2+y-1=0$

$\Rightarrow y=\frac{-1+\sqrt{5}}{2}$

$\therefore f'(y)=\frac{\sqrt{5}+1}{2}, for\ x=1$

54. 00005.00

Sol. $b\cos x+\frac{b\cos 3x}{2}=b\cos x\left(2\cos^2 x-\frac{1}{2}\right)$

$=|\ b\cos x\ |\left|\frac{3}{2}-2\sin^2 x\right|$

$\le \frac{3}{2}|b|\le 3$

$\Rightarrow |b|\le 2 \Rightarrow b=-2,-1,0,1,2.$